

OCR Post-Correction Using Lightweight Large Language Models: A Case Study on the Udmurt Language

Anonymous submission

Abstract

Optical Character Recognition (OCR) systems for low-resource languages frequently exhibit high error rates, primarily due to limited training data and complex morphological structures. This paper addresses the task of OCR post-correction for the Udmurt language, proposing a lightweight yet effective solution based on Large Language Models (LLMs). We introduce a two-stage training pipeline built upon the Qwen2.5-Instruct architecture (1.5B parameters). In the first stage, we apply Supervised Fine-Tuning (SFT) with Low-Rank Adaptation (LoRA) using pairs of noisy OCR output and expert-verified ground truth. This stage achieves a substantial improvement over the commercial ABBYY FineReader baseline, reducing Character Error Rate (CER) by approximately 40% and Word Error Rate (WER) by 71%. To evaluate the generalizability of our approach, we additionally test the trained model on outputs from PaddleOCR-VL-1.6, an open-source multilingual OCR engine, demonstrating its applicability across different OCR systems. In the second stage, we employ Direct Preference Optimization (DPO) to further refine model performance. By leveraging plausible errors generated by the SFT model as rejected responses, we align the model toward preferring correct outputs. This additional stage further reduces the CER to 0.86% (a total relative reduction of 44.5% compared with the ABBYY FineReader baseline). While SFT achieves the best WER, DPO primarily improves character-level accuracy and reduces the proportion of sentences with length mismatches by 16%. Our results demonstrate that a lightweight LLM, fine-tuned and aligned through this pipeline, offers a robust and computationally efficient solution for OCR correction in underrepresented languages.

Introduction

The digitization of textual heritage through Optical Character Recognition (OCR) is a critical step in preserving and making information accessible. While modern OCR systems demonstrate high accuracy for high-resource languages, they often struggle significantly with low-resource languages. This discrepancy is largely due to the scarcity of annotated training data and the complex morphological structures inherent in many underrepresented languages. Consequently, the raw output from standard OCR engines for

such languages frequently contains errors—including mis-recognized characters, omitted words, and erroneous insertions—which severely hinder downstream natural language processing tasks.

Udmurt, a Finno-Ugric language spoken in the Udmurt Republic, exemplifies these challenges. Despite the existence of commercial OCR engines like ABBYY FineReader, the accuracy of automated recognition for Udmurt texts remains suboptimal. Traditional post-processing methods, such as rule-based spell checkers, often fail to correct context-dependent errors or restore morphological endings that are visually ambiguous or completely lost during the recognition phase. Therefore, there is a pressing need for more intelligent, context-aware post-correction mechanisms.

Recent advancements in Large Language Models (LLMs) offer a promising solution to this problem. LLMs have demonstrated remarkable capabilities in understanding context and semantics, allowing them to effectively correct noisy text. However, deploying massive state-of-the-art models for low-resource language tasks can be prohibitively expensive and computationally intensive. Furthermore, standard fine-tuning approaches may not be sufficient to align the model perfectly with the specific nuances of error correction required to outperform established commercial baselines.

In this study, we demonstrate that a lightweight instruction-tuned LLM can outperform a commercial OCR system through post-correction alone, and that preference alignment with synthetic errors improves correction consistency. We leverage the Qwen2.5-Instruct model (1.5B parameters), balancing performance with computational efficiency. Our approach employs a two-stage training strategy. First, we apply Supervised Fine-Tuning (SFT) with Low-Rank Adaptation (LoRA) to adapt the model to the specific task of correcting OCR errors. Second, we utilize Direct Preference Optimization (DPO) to further align the model’s outputs with human-like corrections, specifically targeting issues such as sequence length mismatches. We evaluate our approach against the ABBYY FineReader baseline, demonstrating substantial improvements in both Character Error Rate (CER) and Word Error Rate (WER). Additionally, to assess the robustness of the proposed post-correction method across different OCR systems, we evaluate the model on outputs produced by PaddleOCR-VL-1.6. This allows us to measure whether the learned correction capability generalizes beyond

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the OCR engine used during training.

The remainder of this paper is structured as follows. Section Related work reviews related studies in OCR post-correction and LLM fine-tuning. Section Dataset details the dataset construction, while Section Method describes the SFT and DPO training procedures, and Section Experimental Setup contains the details of the experimental setup. Section Results presents the quantitative and qualitative results of our evaluation. Finally, Section Conclusion discusses the implications of our findings and outlines future research directions.

Related Work

Recent OCR post-correction approaches have moved from rule-based methods and statistical language models toward neural sequence-to-sequence architectures. Character-level neural machine translation models demonstrated that OCR correction can be formulated as a transformation from noisy recognition output to clean text, enabling context-dependent correction beyond manually designed rules (Hämäläinen and Hengchen 2019; Hashmi et al. 2019). More recent approaches employ pretrained Transformer-based models, including mT5 and mBART variants, achieving substantial improvements in OCR correction quality (Hajiali 2023; Aung, Thu, and Oo 2024). Other studies have explored ensemble architectures, hybrid linguistic-neural approaches, and synthetic data generation strategies to improve robustness and reduce dependence on manually annotated OCR corpora (Ramirez-Orta et al. 2022; Schulz and Kuhn 2017; Kashid and Bhattacharyya 2025). However, most existing OCR post-correction methods have been developed for high-resource languages and rely on large parallel OCR datasets.

Applying neural OCR correction methods to low-resource languages remains challenging due to limited annotated data and insufficient representation of these languages in pretrained models. Recent research on low-resource languages has therefore focused primarily on corpus creation and adaptation of multilingual language models rather than OCR post-correction. For Udmurt, several resources have been developed, including monolingual and parallel corpora, as well as inclusion in multilingual benchmarks such as Glot500-c (Arkhangelskiy and Medvedeva 2016; Yankovskaya et al. 2023; ImaniGooghari et al. 2023). Subsequent studies investigated adaptation of pretrained multilingual models for Udmurt and related Uralic languages using approaches such as vocabulary adaptation and language-adaptive pre-training (Isaeva, Astafurov, and Martynov 2025; Downey et al. 2024; Lin et al. 2024).

Recent advances in large language models have also enabled experiments on Udmurt text generation, transliteration, and lexical post-editing tasks (Partanen 2024; Nielsen et al. 2025; Tereshchenko, Hämäläinen, and Myroniuk 2025). Nevertheless, OCR post-correction for Udmurt remains largely unexplored. Although modern OCR systems such as PaddleOCR-VL support Udmurt among their multilingual capabilities, dedicated evaluations and correction methods for Udmurt OCR outputs have not been reported (Cui et al. 2025). To the best of our knowledge, this work is the

first to investigate instruction-tuned lightweight large language models combined with preference optimization for OCR post-correction in this low-resource language.

Dataset

ID	OCR Input	Ground Truth
1	Ымдурьёсыз векчесь, улалысьез дырекъя оло маке вераны турттэменыз по дН-сдтмтэсыз, оло кышкаменыз.	Ымдурьёсыз векчесь, улалалысьез дырекъя оло маке вераны турттэменыз по дйсьтымтэеныз, оло кышкаменыз.
2	Умось-а? — вёзэз ик куири кылйськиз.	Умось-а? — вёзэз ик куара кылйськиз.
3	Тйни вераське тй соип! — иптысьсыз пош ик тэтчиз Рыбин.	Тйни вераське тй соип! — иптысьсыз пош ик тэтчиз Рыбин.
4	Жюген Олексан вуыны кулэ.	Жюген Олексан вуыны кулэ.

Figure 1: Examples of training data pairs showing typical OCR errors and their corresponding ground truth corrections.

The study was conducted on 50 literary texts in the Udmurt language, provided by the Udmurt Federal Research Center of the Ural Branch of the Russian Academy of Sciences (Udmurt FRC UB RAS). A limitation of this work is that the evaluation dataset cannot be publicly released due to institutional data-sharing restrictions. The original data consist of three sets: (1) PDF files of scanned book pages; (2) text files of the books proofread by experts from the research center; and (3) text files generated by the ABBYY FineReader OCR system. Additionally, PaddleOCR-VL-1.6 was applied to the same scanned documents to generate an independent OCR output. These results were not used during training and were reserved for evaluating the generalization ability of the proposed post-correction approach.

For the development and testing of the proposed approach, only the textual data were used. The texts were cleaned by removing extraneous characters and expert markup tags such as <rus>, <eng>, <ger>, <net>, <deu>, and others. These tags indicate passages in languages other than Udmurt (e.g., Russian, German, Tatar, etc.).

Subsequently, sentence-level alignment was performed using the LF Aligner tool. This process yielded a dataset comprising a total of 74,398 sentence pairs. Among these, 26,977 sentences contained recognition errors, while 47,421 sentences were error-free. The error-free sentences were retained in the dataset to ensure that the model learns not only to correct errors but also to preserve accurate OCR outputs.

The resulting dataset was split into training, validation and test subsets in an 80%–10%–10% ratio. Table 1 presents the size of each subset along with the corresponding Character Error Rate (CER) and Word Error Rate (WER) metrics for both ABBYY FineReader and PaddleOCR-VL-1.6 on the test set.

Figure 1 presents example training instances from the dataset, illustrating typical OCR errors and their corresponding ground truth corrections. Each row shows a pair of sentences: the first line contains the expert-verified ground truth,

Subset	Rows	CER (%)		WER (%)	
		ABBYY FineReader	PaddleOCR	ABBYY FineReader	PaddleOCR
Train	59,499	1.54	—	8.27	—
Validation	7,449	1.56	—	8.34	—
Test	7,450	1.55	3.86	8.33	17.67
Total	74,398	1.54	—	8.28	—

Table 1: Dataset statistics by subset.

and the second line contains the corresponding OCR output from ABBYY FineReader. The examples demonstrate various types of errors commonly encountered in Udmurt OCR output, including character substitutions, diacritic omissions, and complex character misrecognition patterns. These samples reflect the diverse error patterns that the model must learn to address.

For the second-stage preference alignment, no additional human preference annotations were collected. Instead, the preference pairs were automatically constructed from the intermediate SFT model. The expert-verified ground-truth sentences were used as chosen responses, while rejected responses were generated by the SFT model itself. These generated outputs represent plausible but imperfect OCR corrections, allowing the DPO stage to learn from realistic model-specific errors without requiring additional manual annotation.

Method

We propose a two-stage training pipeline based on a lightweight pre-trained Large Language Model (LLM), comprising sequential stages of supervised fine-tuning (SFT) and preference alignment via Direct Preference Optimization (DPO).

The first stage, SFT, is designed to adapt the general-purpose base model to the specific downstream task of OCR post-correction. At this phase, we utilize a labeled dataset consisting of pairs of noisy OCR output—obtained using the FineReader engine—and the corresponding ground-truth text, which has been verified by linguistic experts.

The second stage introduces preference alignment utilizing the DPO algorithm. As a modern alternative to classical reinforcement learning from human feedback (RLHF), DPO offers a simplified yet effective framework that obviates the need for training a separate reward model. Instead, this method operates directly on pairwise preference data, comprising chosen and rejected responses. To construct the rejected responses, we leverage the SFT model checkpoint obtained after the first training epoch. At this intermediate stage, the SFT model has already acquired foundational knowledge of the Udmurt language and the OCR correction task. Nonetheless, it produces erroneous yet linguistically plausible outputs—a property we deliberately exploit to construct quality rejected responses for DPO training. The original expert-verified ground-truth sentences serve as the chosen responses, thereby guiding the model toward preferring accurate outputs over its own early-stage mistakes.

Unlike conventional DPO settings that rely on human preference annotations, we construct rejected responses automatically from intermediate SFT checkpoints, exploiting their plausible but imperfect corrections.

Experimental Setup

In this study, the lightweight instruction-tuned language model Qwen2.5-1.5B-Instruct serves as the foundation model for OCR post-correction. While ABBYY FineReader is used as the primary OCR engine for constructing the training corpus and for the main evaluation, PaddleOCR-VL-1.6 is additionally included as a secondary baseline to assess the robustness and generalization capability of the proposed post-correction approach across different OCR systems.

ABBYY FineReader: As a commercial state-of-the-art OCR engine¹, FineReader provides the primary baseline for our experiments. It combines document layout analysis, optical character recognition, and language-aware post-processing. The raw OCR output generated by FineReader is used as the input to the proposed correction model. The corresponding CER and WER values reported in Section represent the baseline performance before language model post-correction.

PaddleOCR-VL-1.6: As an additional reference system, we evaluate PaddleOCR-VL-1.6, an open-source multilingual OCR framework supporting 109 languages, including Udmurt (Cui et al. 2025). Unlike FineReader, PaddleOCR is based on deep neural architectures and is designed for end-to-end document understanding. Including this OCR engine allows us to evaluate whether the proposed post-correction model generalizes beyond the OCR system used during training. The recognition accuracy of both OCR systems is compared in Section .

All experiments were conducted in the Kaggle Notebook environment using Python 3.12, PyTorch 2.10 with CUDA 12.8, the Hugging Face Transformers library, PEFT, and TRL. Training and inference were performed on an NVIDIA Tesla T4 GPU with 15 GB of memory using automatic device mapping and mixed-precision (bf16) computation.

The tokenizer of Qwen2.5-1.5B-Instruct was used without modification. No additional vocabulary entries were introduced, and the tokenizer was neither retrained nor extended with Udmurt-specific tokens. Instead, all Udmurt characters were preserved during preprocessing, allowing the model to

¹ABBYY FineReader 16. <https://pdf.abbyy.com/>

leverage its original multilingual vocabulary together with task-specific fine-tuning.

Evaluation was performed using Character Error Rate (CER) and Word Error Rate (WER), which served as the primary quantitative metrics throughout the experiments. Both metrics were computed using the jiwer library after applying the same text preprocessing pipeline used during corpus construction. The preprocessing included removal of dataset-specific markup and unsupported symbols, normalization of consecutive whitespace into a single space, normalization of quotation marks, restoration of spaces after sentence-ending punctuation where required, and removal of line breaks introduced by hyphenation. Sentence segmentation was performed using the razdel library with additional splitting at semicolons. The original letter case was preserved throughout the evaluation, and no Unicode normalization (e.g., NFC or NFKC) was applied. WER was computed using the default whitespace-based tokenization provided by jiwer, whereas CER was calculated at the character level.

Training Details

SFT Stage: Parameter-efficient fine-tuning was performed using the LoRA method with rank $r = 16$, scaling factor $\alpha = 32$, and dropout rate of 0.05. LoRA adapters were inserted into the attention projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and the feed-forward projection layers (`gate_proj`, `down_proj`), while all remaining model parameters were kept frozen. Training was implemented using the SFTTrainer from Hugging Face TRL.

The instruction-tuning format is illustrated in Figure 2. Each training sample consists of a system instruction defining the OCR correction task, a user message containing the OCR output, and an assistant response containing the corresponding ground-truth sentence.

The model was trained for two epochs using a per-device batch size of one with gradient accumulation over four steps (effective batch size of four). The AdamW optimizer was employed with an initial learning rate of 5×10^{-4} , cosine learning-rate scheduling, and 1000 warm-up steps. Validation and checkpointing were performed every 500 optimization steps while mixed-precision (`bfloat16`) training was enabled throughout the optimization process.

```

### system
You are an expert in correcting optical character recognition (OCR) errors in Udmurt language texts. Return only the corrected text without comments.

### user
Цехлэн секыт шокамеэ Алялы азьло кыче ке бадзым пйшурлэн жуштэмез кадь кылйськылиз.

### assistant
Цехлэн секыт шокамеэ Алялы азьло кыче ке бадзым пйшурлэн жуштэмез кадь кылйськылиз.

```

Figure 2: Instruction-tuning dataset sample for SFT.

DPO Stage: Preference optimization was performed using the DPOTrainer implementation from Hugging Face TRL. The corresponding training data format is illustrated in Figure 3. To construct rejected responses, the SFT model generated three candidate corrections for each OCR sentence

using stochastic decoding with `temperature=0.9` and `num_return_sequences=3`. Only samples for which at least one generated candidate achieved a Character Error Rate (CER) between 1% and 30% were retained. This filtering procedure resulted in a preference dataset containing approximately 22,000 training examples.

To preserve the linguistic knowledge acquired during supervised fine-tuning while optimizing human preferences, we adopted a hybrid objective consisting of 80% DPO loss and 20% SFT loss with a preference scaling parameter $\beta = 0.01$.

The DPO stage was performed for one epoch using a per-device batch size of one with gradient accumulation over eight steps (effective batch size of eight). The learning rate was reduced to 1×10^{-6} and optimized using cosine scheduling with 300 warm-up steps.

```

### system
You are an expert in correcting optical character recognition (OCR) errors in Udmurt language texts. Return only the corrected text without comments.

### user
Цехлэн секыт шокамеэ Алялы азьло кыче ке бадзым пйшурлэн жуштэмез кадь кылйськылиз.

### assistant
Chosen: Цехлэн секыт шокамеэ Алялы азьло кыче ке бадзым пйшурлэн жуштэмез кадь кылйськылиз.
Rejected: Цехлэн секыт шокамеэ Алялы азьло кычез дышетэмез кадь кылйськылиз.

```

Figure 3: Instruction-tuning dataset sample for DPO.

Results

The CER and WER quality metric values obtained on the test dataset are presented in Table 2 and Table 3.

The base Qwen2.5-1.5B-Instruct model, without task-specific adaptation, performs substantially worse than the original OCR output (Table 2), yielding a CER of 9.77% and a WER of 23.32%. This behaviour is expected for three reasons. First, Udmurt is severely underrepresented in the multilingual pre-training data of general-purpose LLMs. Second, OCR post-correction requires character-level precision and faithful copying of already correct text, whereas instruction-tuned language models are primarily optimized for semantic generation. Consequently, the base model frequently introduces unnecessary modifications and fails to reliably preserve the original OCR output. These observations further motivate the need for task-specific adaptation through supervised fine-tuning and preference optimization.

The evaluation was conducted on two types of OCR outputs: ABBYY FineReader, which was used for training the post-correction model, and PaddleOCR-VL-1.6, which represents an independent OCR system for testing the generalization capability of the proposed approach.

The raw PaddleOCR output exhibits significantly higher error rates compared to ABBYY FineReader, with CER of 3.86% and WER of 17.67%, which is approximately twice as high in terms of CER. This performance gap highlights the challenges faced by general-purpose open-source OCR systems when applied to low-resource languages with complex morphology such as Udmurt, and underscores the necessity of specialized post-correction approaches.

Model	CER (%)		WER (%)	
	Value	Δ	Value	Δ
ABBY FineReader	1.55	–	8.33	–
Qwen2.5-Instruct 1.5B (base)	9.77	+530.3	23.32	+180.0
+ SFT	0.93	-40.0	2.45	-70.6
+ SFT + DPO	0.86	-44.5	2.54	-69.5

Table 2: OCR post-correction results on ABBYY FineReader.

Model	CER (%)		WER (%)	
	Value	Δ	Value	Δ
PaddleOCR-VL-1.6	3.86	–	17.67	–
+ SFT (trained on ABBYY FineReader outputs)	3.16	-18.1	10.46	-40.8
+ SFT + DPO (trained on ABBYY FineReader outputs)	3.28	-15.0	11.60	-34.4

Table 3: Ablation study on PaddleOCR-VL-1.6 (cross-OCR generalization).

For the primary benchmark based on ABBYY FineReader (Table 2), supervised fine-tuning substantially improves the OCR output, reducing CER from 1.55% to 0.93% (40.0%) and WER from 8.33% to 2.45% (70.6%). Applying DPO after SFT further decreases CER to 0.86%, corresponding to an additional relative improvement of 7.5% over the SFT model, while WER remains largely unchanged.

Figure 4 presents several examples of corrections made by the resulting models on ABBYY FineReader output. It can be observed that the application of DPO minimized the discrepancy in sequence lengths between the ground truth and the LLM responses. DPO reduced the proportion of sentences containing length mismatch errors—caused by erroneous deletions or insertions by the SFT model—by 16%. The examples in the third and fourth rows of Figure 4 illustrate this improvement.

Similarly, Figure 5 demonstrates the model’s performance on PaddleOCR-VL-1.6 output, showing consistent error correction capabilities across different OCR systems.

When applied to PaddleOCR-VL-1.6 outputs (Table 3), the proposed model successfully transfers the correction knowledge learned from ABBYY FineReader data. The SFT stage reduces the CER from 3.86% to 3.16% and the WER from 17.67% to 10.46%, corresponding to relative improvements of 18.1% and 40.8%, respectively. These results demonstrate that the model learns general OCR correction patterns rather than memorizing only the error distribution of the training OCR system.

The additional DPO alignment stage provides a smaller improvement for PaddleOCR-VL-1.6 outputs compared to ABBYY FineReader. After DPO training, the CER slightly increases to 3.28% and the WER to 11.60%. This behavior can be explained by the difference between the error distributions of the two OCR systems: the preference data used for DPO was generated from ABBYY FineReader-based errors, and therefore the learned preferences are better aligned with this OCR source. Nevertheless, the SFT model maintains a substantial improvement over the original PaddleOCR-VL-

1.6 output, confirming the robustness of the proposed post-correction approach.

Discussion

Additional analyses, including evaluation on error-only subsets, overcorrection rate, error-type distribution (substitutions, insertions, and deletions), sentence length mismatch statistics, inference efficiency, and qualitative failure cases, are provided in the supplementary material. These analyses further characterize the behavior of the proposed models without changing the main conclusions of this work.

The experimental results indicate that the two-stage SFT+DPO pipeline provides the best character-level accuracy on ABBYY FineReader outputs, reducing CER from 0.93% after SFT to 0.86%. Although the improvement is relatively modest, it is consistent with the objective of DPO. Since the preference dataset is constructed entirely from ABBYY OCR outputs, DPO learns to prefer corrections that better match the characteristic error patterns produced by this OCR engine, resulting in additional refinement beyond supervised fine-tuning.

At the same time, the cross-OCR experiments reveal a limitation of this approach. While SFT improves the performance of both ABBYY FineReader and PaddleOCR-VL, DPO slightly degrades the results on PaddleOCR-VL. This behaviour suggests that the preference optimization stage becomes partially specialized to ABBYY-specific error distributions. Because PaddleOCR generates a different combination of insertion, deletion, and substitution errors, the learned preferences transfer less effectively to another OCR system.

A similar observation can be made for WER. Although DPO further reduces CER, it produces a slight increase in WER compared with SFT alone. This indicates that DPO primarily improves character-level corrections rather than complete word restoration. Minor character modifications may reduce the number of character errors while still producing an incorrect lexical form, resulting in a small increase

ID	ABBYY FineReader	Ground Truth	SFT	SFT + DPO
1	Озыен, тй госу- дарственной ила- изз бд быдэстэ?	Озыен, тй го- сударственной планэз өд быд- эстэ?	Озыен, тй госу- дарственной ин- тые бд быдэстэ?	Озыен, тй го- сударственной планэз өд быд- эстэ?
2	Вылтйз век- чи ейен куро вөлдйзы.	Вылтйз век- чи сийен куро вөлдйзы.	Вылтйз век- чи сийен куро вөлдйзы.	Вылтйз век- чи ейен куро вөлдйзы.
3	Планёркмчс дыръя но юн- ме шчрыюь кесяське, участо- къяёслэсь кивил тй с тышкаське.	Планёркаос дыръя но юн- ме шорысь уча- кесяське, участо- къяёслэсь кивалтйсьёссэс тышкаське.	Планёрщике дыръя но юнме шорысь кесясь- ке, участокъяёсл- эсь кивалтйсьёс сярысь сюл- маське.	Планёркаческой дыръя но юнме шорысь кесясь- ке, участокъяёсл- эсь кивалтйс тышкаське.
4	Тодэ лытэм кочыш, кытысь Нөлвыт усс.	Тодэ вылэм кочыш, кытысь йөлвыл усе.	Тодэ лыктэм көшкемыт усе.	Тодэ лытэм кочыш, кытысь йөлвыл усе.

Figure 4: Comparison of OCR outputs for ABBYY FineReader.

ID	PaddleOCR	Ground Truth	SFT	SFT + DPO
1	Озымен, тй го- сударственной планэз од быд- эстэ?	Озыен, тй го- сударственной планэз өд быд- эстэ?	Озыен, тй го- сударственной планэз өд быд- эстэ?	Озымен, тй го- сударственной планэз өд быд- эстэ?
2	Вылтйз векчи сий- ен куро волдй- зы.	Вылтйз век- чи сийен куро вөлдйзы.	Вылтйз век- чи сийен куро вөлдйзы.	Вылтйз век- чи сийен куро вөлдйзы.
3	Планёркаос дыръя но юнме шорысь кесяське, участокъяёслэсь кивалтйсьёссэс тышкаське.	Планёркаос дыръя но юнме шорысь кесяське, участокъяёслэсь кивалтйсьёссэс тышкаське.	Планёркаос дыръя но юнме шорысь кесяське, участокъяёслэсь кивалтйсьёссэс тышкаське.	Планёркаос дыръя но юнме шорысь кесяське, участокъяёслэсь кивалтйсьёссэс тышкаське.
4	Тодэ вылэм кочыш кытысь йолвыл усе.	Тодэ вылэм кочыш кытысь йөлвыл усе.	Тодэ вылэм кочыш кытысь йөлвыл усе.	Тодэ вылэм кочыш кытысь йөлвыл усе.

Figure 5: Comparison of OCR outputs for PaddleOCR-VL-1.6.

in word-level error rate.

Overall, the results suggest that SFT provides a more robust and OCR-agnostic correction model, while DPO acts as a task-specific refinement stage that benefits the OCR engine used to construct the preference dataset. A promising direction for future work is the construction of mixed preference datasets combining outputs from multiple OCR systems. Such a strategy could expose DPO to a wider variety of OCR error distributions and improve both robustness and cross-engine generalization.

Conclusion

In this paper, we addressed the challenge of post-processing OCR output for the low-resource Udmurt language by proposing a pipeline based on lightweight Large Language Models. By leveraging the Qwen2.5-Instruct model with 1.5 billion parameters, we demonstrated that efficient fine-tuning can achieve strong performance for this low-resource setting without the computational overhead of massive models. Our two-stage approach—consisting of Supervised Fine-Tuning (SFT) followed by Direct Preference Optimization

(DPO)—proved highly effective in transforming a general-purpose LLM into a specialized tool for OCR correction.

The experimental results showed a substantial improvement over the commercial ABBYY FineReader baseline (Table 2). The SFT stage alone reduced the Character Error Rate (CER) by 40% and the Word Error Rate (WER) by 71%. Furthermore, the incorporation of DPO provided additional gains, achieving a total CER reduction of 44.5%. Qualitative analysis revealed that DPO specifically addressed structural weaknesses of the SFT model, such as sequence length mismatches caused by erroneous insertions or deletions, reducing the frequency of such errors by 16%. Additionally, evaluation on PaddleOCR-VL-1.6 outputs (Table 3) demonstrated the generalization capability of the proposed approach. The SFT model reduced the CER by 18.1% and the WER by 40.8% compared to the original PaddleOCR-VL-1.6 recognition results, confirming that the learned correction patterns are transferable across different OCR systems.

Despite these promising results, the proposed approach has several limitations. Although the supplementary material provides additional analyses including overcorrection rates, error-type statistics, and inference efficiency, the evaluation remains limited to two OCR systems. Future work should investigate a broader range of OCR engines and more diverse document collections. The experiments also were conducted on sentence-level post-correction. Exploiting document-level context may further improve correction quality for ambiguous OCR errors. Both the supervised fine-tuning and Direct Preference Optimization (DPO) stages were constructed using OCR outputs generated by ABBYY FineReader. Although the model demonstrated good generalization to PaddleOCR-VL-1.6, its performance on outputs produced by substantially different OCR engines remains to be investigated. Furthermore, the proposed method operates solely on recognized text and does not exploit visual information from the original document, preventing it from resolving errors that depend on page layout, typography, or image context. Finally, the training corpus consists exclusively of printed Udmurt documents, and the proposed approach has not yet been evaluated on handwritten or severely degraded historical materials.

This study highlights the potential of using aligned, lightweight LLMs for digital humanities and language preservation tasks involving underrepresented languages. The ability to correct complex morphological errors using a model with only 1.5 billion parameters suggests that similar methodologies can be deployed in resource-constrained environments. Future work will investigate document-level OCR post-correction by incorporating broader textual context beyond individual sentences, evaluate the proposed approach on outputs produced by a wider range of OCR engines, and extend the framework to additional low-resource languages. Another promising direction is the development of more diverse preference datasets together with improved mixed DPO objectives that better balance supervised learning and preference optimization.

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